

Mass, Inertia, and Persistence Compression

The Deformation-Resistance Role of Particle Identity under Field-Carrier Mediation

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Abstract

This paper derives the minimal mass/inertia role within the realized-physics arc. Previous results established that realized persistence cannot be physically null, that standing-bearing persistence localizes, that localized standing induces continuation-neighborhood domains, that stable comparison supplies the metric role of a quotient comparison carrier, that perturbation transport is bounded by continuation-comparison constraints, that fields are persistence carriers, that particles are localized persistence classes, and that gauge structure is boundary-fixed presentation equivalence. The present paper asks what is required for a particle persistence class to remain the same under admissible deformation and transport-state change. It proves that identity-preserving deformation requires a stable compression-response profile: the admissible carrier/restoration/transport adjustment required to preserve standing identity, represented as a gauge-compatible, boundary-fixed, quotient-stable comparison class.

The main result is a mass-inertia role induction theorem. In a fixed admissible realized persistence regime with field-carrier assignment Φ_D , particle persistence classes Π_D , gauge-role structure $\mathcal{G}_D$, bounded perturbation transport, and quotient comparison carrier C_D , any admissible deformation family that preserves particle identity induces a partial mass-role profile

$$\mu_D(\Pi) : \Delta_\Pi \rightarrow C_D$$

and an inertia-role profile obtained by restriction to transport-state deformations,

$$\iota_D(\Pi) = \mu_D(\Pi)|_{\Delta_\Pi^{\text{tr}}}.$$

This is the minimal physical role later occupied by mass parameters and inertial response: operative resistance of a localized particle persistence class to standing-preserving deformation, not merely a descriptive burden ledger. Numerical mass spectra, Higgs/Yukawa mechanisms, QCD mass generation, relativistic mass-energy relations, inertial/gravitational equivalence, stress-energy coupling, mass-shell dynamics, dispersion relations, and field equations are not assumed. They are later specializations of the deformation-resistance role isolated here.

1 Introduction

The realized-physics arc has reconstructed familiar physical roles in a fixed order. The first paper establishes the non-null floor of physical realization: realized persistence cannot be reconstruction-null inside its own admissible continuation regime [1]. The second paper derives bounded standing-localization as the origin of physical locality [2]. The third derives continuation-neighborhood domains as the origin of local-domain structure [3]. The fourth derives the metric role as quotient-stable comparative representation among local continuation domains [4]. The fifth derives bounded perturbation transport as the finite-propagation-constraint role [5]. The sixth derives fields as

persistence carriers [6]. The seventh derives particles as localized persistence classes [7]. The eighth derives gauge structure as boundary-trace-fixed presentation equivalence [8].

In compact form, the chain is

$$\begin{aligned} \text{RealPers}_D &\Rightarrow \text{NonVacRec}_D \Rightarrow [B_D]_{\equiv_B} \\ &\Rightarrow (\text{Adj}_D, \mathcal{N}_D, \text{LCD}_D) \Rightarrow C_D \Rightarrow \text{BoundTrans}_D \\ &\Rightarrow \Phi_D \Rightarrow \Pi_D \Rightarrow \mathcal{G}_D. \end{aligned}$$

The present analysis begins at the final three terms. A particle is now a localized persistence class supported by field-carrier mediation and stabilized under boundary-fixed gauge presentation. The next physical question is:

If a particle persistence class remains identifiable while its carrier configuration is deformed, transported, restored, or re-presented, what fixes the response profile of that identity-preserving change?

The answer is the mass/inertia role. Mass and inertia first enter realized physics as the gauge-compatible compression-response profile by which a localized particle persistence class resists standing-threatening deformation under field-carrier mediation. The profile is operative: modifying it changes which deformations and transport-state changes preserve the class.

1.1 Theorem claim and physical role

The theorem-level claim is

$$(\Phi_D, \Pi_D, \mathcal{G}_D, \text{BoundTrans}_D, C_D) \wedge \text{DefAdm}_D \wedge \text{IdRet}_D \Rightarrow \text{MassInertiaRole}_D.$$

The physical-role claim is that MassInertiaRole_D is the minimal deformation-resistance role occupied by later mass parameters and inertial response laws. The result does not define mass as primitive substance, primitive scalar value, field energy, gauge presentation, or measurement output. It defines the role a mass parameter must play if it is to be physically meaningful inside the derived structure: it must represent how a particle persistence class preserves standing identity under admissible deformation and transport-state change.

This role-first treatment is physical rather than merely bookkeeping. Familiar mass quantities are special representations of a prior physical role: identity-preserving resistance under deformation. A descriptive burden records how a deformation is classified; a mass/inertia role constrains which standing-preserving deformations and transport-state changes remain admissible. A number written as m , a rest-mass label, a Yukawa eigenvalue, an effective running mass, or a stress-energy contribution becomes mass-relevant only insofar as it tracks that role under the admissible operations of the regime.

Paper	Structural result	Physical role
1	NonVacRec_D	physical realization cannot be null
2	$[B_D]_{\equiv_B}$	origin of physical locality
3	$(\text{Adj}_D, \mathcal{N}_D, \text{LCD}_D)$	origin of local-domain organization
4	$C_D = \text{CompWit}_D / \sim_D$	origin of metric comparison role
5	BoundTrans_D	origin of finite-propagation constraint
6	Φ_D	origin of field-carrier mediation
7	$\Pi_D = [\kappa_U]_{\equiv_\Pi}$	origin of particle identity
8	$\mathcal{G}_D$	origin of gauge structure
9	(μ_D, ι_D)	origin of mass/inertia role

1.2 Scope

The paper derives the minimal mass/inertia role. It does not derive electron, quark, neutrino, gauge-boson, or Higgs masses; Higgs vacuum structure; Yukawa matrices; QCD confinement; binding energy; inertial/gravitational equivalence; stress-energy tensors; mass-shell relations; dispersion relations; relativistic energy-momentum dynamics; or field equations. Those are later specializations. What is derived here is the prior physical role: a particle persistence class can undergo identity-preserving deformation only when the regime supplies a quotient-stable compression-response profile.

2 Dependencies and imported structure

The paper imports the first eight results in the realized-physics arc and selected source machinery concerning standing margin, physical constants, renormalization, gauge compatibility, and fixed-domain discipline. The dependency structure is summarized in Table 1.

Table 1: Imported structure for the mass/inertia-role derivation.

Imported result	Role in the present paper
Non-vacuous realized persistence	A mass role cannot attach to null realization; it requires non-null standing-bearing persistence.
Bounded standing-localization	Deformation response must be localized; it cannot be assigned to a global or diffuse object without losing standing verdicts.
Continuation-neighborhood domains	Deformation and transport-state change occur over local continuation domains, not abstract points.
Quotient comparison carrier C_D	Compression response is compared through quotient-stable comparison classes rather than raw scalar import.
Bounded perturbation transport	Inertia is tied to bounded changes in transport/restoration profiles.
Field-carrier mediation Φ_D	The deformation response is carried by field-carrier mediation, not primitive substance.
Particle persistence class Π_D	Particle identity supplies the class whose standing-preserving deformation response is measured.
Gauge-role structure $\mathcal{G}_D$	Mass/inertia profiles must be gauge-compatible and boundary-trace fixed, not presentation artifacts.
Standing-margin compression	Repair-minus-burden structure supplies the role ancestor for compression response.
Physical constants and RG sources	Mass values, mass ratios, running masses, and effective masses are role-dependent representations, not primitive same-domain scalars.
Standard Model classification sources	Higgs/Yukawa mass generation, gauge-boson masses, and mixing bases are later specializations of the mass role.
AMetric boundary discipline	Primitive scalar mass assignments, hidden mass reservoirs, default metric scales, and arbitrary extraction selectors are inadmissible.

Imported result		Role in the present paper
Anchor–tensor–skin	disci-	Mass labels or numbers are skin unless they alter standing-bearing deformation response.
pline		

The key source bridge is standing-margin compression. Earlier standing-localization work identifies the standing margin as a retained repair/restoration capacity minus perturbation burden. Paper 9 generalizes this local viability structure to particle persistence classes: mass/inertia is not the margin itself, but the stable compression-response profile required when admissible deformation threatens the class while identity is retained.

3 Definitions and type discipline

Definition 3.1 (Particle persistence class). A particle persistence class is the localized carrier-mediated identity role derived previously,

$$\Pi_D = [\kappa_U]_{\equiv \Pi},$$

where κ_U is a localized carrier configuration and $\equiv \Pi$ preserves standing signature, boundary/classification witness, restoration profile, quotient comparison, identity-memory, local-domain role, and admitted transport role. The class Π_D is the identity target for the mass/inertia role.

Definition 3.2 (Gauge-compatible carrier presentation). A carrier presentation of a particle class is gauge-compatible when boundary-fixed presentation equivalence preserves the same carrier-mediated particle content. A mass/inertia profile is admissible only if it is invariant under gauge-role transitions preserving the same Π_D . Thus mass/inertia cannot be a presentation artifact.

Definition 3.3 (Admissible deformation). An admissible deformation is a fixed-domain modification witness

$$\delta_D : \kappa_U \rightsquigarrow \kappa'_V$$

that changes, stresses, or threatens at least one of the standing signature, boundary/classification witness, identity-memory, restoration profile, perturbation-burden profile, transport profile, quotient comparison class, or gauge presentation of a localized carrier configuration. The deformation remains within the admissible realized regime; cross-domain reset is not deformation of the same class.

Definition 3.4 (Standing-preserving deformation). A deformation $\delta_D : \kappa_U \rightsquigarrow \kappa'_V$ is standing-preserving for the particle class Π iff

$$\kappa_U \equiv \Pi \kappa'_V.$$

The family of admitted standing-preserving deformations of Π is denoted Δ_Π . A deformation in Δ_Π may change the local carrier configuration, transport profile, or presentation, but it does not destroy the particle persistence class.

Definition 3.5 (Compression response). A compression response is the admissible carrier/restoration/transport adjustment required to preserve the same particle persistence class under an admissible deformation. In this sense, “compression” is not metaphorical squeezing and not a scalar mass law. It is the operative response by which a localized persistence class absorbs,

redirects, or compensates deformation burden while retaining standing identity. Its comparison class is written

$$\text{cmp}(\delta, \Pi) \in C_D.$$

The notation does not assert a scalar magnitude. The compression response may be partially ordered, direction-sensitive, envelope-dependent, or neutral relative to the admitted comparison carrier.

Definition 3.6 (Mass-role profile). The mass-role profile of a particle persistence class Π is the partial map

$$\mu_D(\Pi) : \Delta_\Pi \rightharpoonup C_D, \quad \mu_D(\Pi)(\delta) = [\text{cmp}(\delta, \Pi)].$$

It assigns to each admitted standing-preserving deformation its quotient-stable compression-response class. It is partial because not every formal deformation is admissible, comparable, or class-preserving.

Definition 3.7 (Inertia-role profile). Let $\Delta_\Pi^{\text{tr}} \subseteq \Delta_\Pi$ be the subfamily of standing-preserving deformations that change the bounded transport/restoration profile of Π . The inertia-role profile is

$$\iota_D(\Pi) = \mu_D(\Pi)|_{\Delta_\Pi^{\text{tr}}}.$$

Thus inertia is the transport-facing manifestation of the broader mass role: the resistance profile associated with identity-preserving changes of transport state.

Definition 3.8 (Mass-neutral response). A particle class is response-neutral relative to a deformation family $\Delta_0 \subseteq \Delta_\Pi$ iff $\mu_D(\Pi)(\delta)$ is neutral or vanishing for all $\delta \in \Delta_0$. Neutral response is not nullity. A class may have a response-neutral profile relative to a selected deformation family while still carrying localization, field-carrier, gauge, transport, and particle-identity roles.

4 Operational status of the compression response

The compression-response profile is not a descriptive burden ledger. It is a role structure: changing it changes which deformations, transport-state changes, and restoration operations preserve the same particle persistence class. This distinction separates mass/inertia from bookkeeping. A descriptive burden may record that a deformation occurred; a mass/inertia role constrains how identity is preserved through that deformation.

Proposition 4.1 (Deformation-mediation). *If admissible modification of $\mu_D(\Pi)$ changes the transport/restoration behavior required to preserve Π , then $\mu_D(\Pi)$ is not descriptive classification but operative deformation mediation.*

Proof. The profile $\mu_D(\Pi)$ is defined over standing-preserving deformations. If modifying the profile changes which carrier/restoration/transport adjustments preserve Π , then the profile participates in the admissible preservation of the particle class. A purely descriptive classification could vary without changing the transport/restoration organization of the regime. Therefore a profile whose admissible modification changes identity-preserving deformation behavior is an operative mass/inertia role rather than bookkeeping. $\square$

Remark 4.2 (Neutral response is not nullity). A response-neutral mass profile relative to a deformation family does not remove physical content. It states that the compression-response

class is neutral for that family. The particle class may still possess localized standing, field-carrier mediation, gauge role, bounded transport, boundary witnesses, and particle identity. This distinction prevents the theorem from implying that every physically realized class has a nonzero scalar rest mass.

5 Failure of primitive mass-source candidates

The first step is negative. Mass and inertia cannot be imported as primitive labels before the deformation-response role is fixed.

Theorem 5.1 (No primitive scalar mass assignment). *A scalar mass assignment m introduced before particle identity, field-carrier mediation, gauge compatibility, admissible deformation, and quotient-stable comparison are fixed does not by itself carry a mass role. Scalar mass parameters become later representations of the deformation-response profile $\mu_D(\Pi)$. If m represents that profile, it is a representation of the derived role. If it does not, it is skin, selector import, or bookkeeping.*

Proof. A scalar mass assignment can be written before a regime specifies what is being deformed, what identity is retained, which carrier mediates the deformation, which presentation changes are gauge-equivalent, and how deformation burden is compared. Such an assignment cannot distinguish a physical mass role from unit convention, extraction scheme, coordinate normalization, or label. If it changes under gauge-equivalent presentation, it is not physical mass. If it remains invariant but does not alter standing-preserving deformation response, it is inert notation. Only when the scalar mass assignment tracks $\mu_D(\Pi)$ does it represent the mass role. Hence scalar mass parameters are later representations of the deformation-response role, not primitive sources of that role. $\square$

Lemma 5.2 (Substance-amount insufficiency). *A primitive amount of substance cannot ground the mass role unless it determines standing-preserving deformation response for a particle persistence class. If it does determine that response, the physical mass role is the response profile rather than primitive substance.*

Proof. A substance amount without a deformation-response rule does not say how a particle class retains standing identity when stressed, transported, compressed, or restored. It can be increased, decreased, or relabeled without specifying admissible persistence. If the alleged substance amount fixes such response, then its physically relevant content is precisely the compression profile μ_D . The primitive substance picture is therefore unnecessary for the mass role. $\square$

Lemma 5.3 (Energy-ledger insufficiency). *An energy or transport-capacity ledger does not by itself define mass or inertia unless it fixes the standing-preserving deformation response, persistence stabilization, and restoration compatibility of a particle persistence class.*

Proof. Energy-like ledgers may later measure continuation capacity, work, stress, excitation, interaction budget, or transport capacity. But a ledger alone does not determine the identity-preserving deformation response of a particle class, the persistence stabilization required to retain that class, or the restoration compatibility needed to distinguish admissible deformation from identity loss. It also does not distinguish gauge presentation artifact from gauge-invariant compression role. The mass role is therefore not obtained by naming an energy ledger. It is obtained only when the ledger represents the compression-response profile μ_D for a fixed class Π . $\square$

Lemma 5.4 (Gauge-presentation artifact collapse). *A difference between gauge-equivalent presentations cannot constitute a mass difference unless boundary traces distinguish a different deformation-response role.*

Proof. Gauge-equivalent presentations preserve the same carrier-mediated particle content. Counting a presentation difference as a mass difference without a boundary-trace distinction duplicates physical content across representations. Conversely, if the boundary traces and deformation witnesses distinguish a different compression profile, the presentations are no longer merely gauge-equivalent relative to the mass role. Hence gauge presentation alone cannot carry mass. $\square$

6 Compression response and mass-role induction

The positive construction begins when a deformation threatens a particle persistence class while the class is nevertheless retained. The retained class must absorb, redirect, or compensate the deformation burden through its carrier/restoration/transport structure. This is compression response in the operational sense: the admissible adjustment required to keep the same persistence class standing-identical.

Proposition 6.1 (Compression response requirement). *If an admissible deformation δ threatens a particle persistence class Π while Π remains standing-identical, then a compression response must be assigned to the field-carrier/restoration/transport structure preserving Π .*

Proof. If δ does not threaten any standing-bearing component, it is irrelevant to the class. If it threatens the class and the class is not retained, it is not a standing-preserving deformation. The remaining case is identity retention under threat. The regime must then account for how the threatened standing signature, boundary witness, transport profile, restoration profile, and quotient class remain preserved. That account is not a label; it is an admissible adjustment of the carrier/restoration/transport structure. This required adjustment is the compression response. $\square$

The source-level standing-margin form $\Sigma = \tilde{R} - D$ is a role ancestor of this result, not a literal mass equation and not a derived scalar law. It supplies a structural pattern: persistence under burden requires retained restorative capacity, and the relevant physical role is the response by which deformation burden is carried without loss of standing identity.

Proposition 6.2 (Mass role from standing-preserving deformation). *For any particle persistence class Π , the admitted family of standing-preserving deformations Δ_Π induces a quotient-stable mass-role profile*

$$\mu_D(\Pi) : \Delta_\Pi \rightarrow C_D.$$

Proof. For each $\delta \in \Delta_\Pi$, the deformation preserves Π while changing or stressing some carrier, restoration, boundary, transport, or comparison component. By the compression response requirement, there is a compression response associated with preserving Π . By Paper 4, admissible comparison of such responses must be represented in the quotient comparison carrier C_D , not by arbitrary scalar import. By Paper 8, gauge-equivalent presentations cannot alter the physical response. Therefore each admitted deformation determines a quotient-stable comparison class $[\text{cmp}(\delta, \Pi)]$. The partial assignment $\delta \mapsto [\text{cmp}(\delta, \Pi)]$ is $\mu_D(\Pi)$. $\square$

Proposition 6.3 (Inertia role). *If a standing-preserving deformation changes the bounded transport/restoration profile of a particle class, the mass-role profile restricts to an inertia-role profile*

$$\iota_D(\Pi) = \mu_D(\Pi)|_{\Delta_\Pi^{\text{tr}}}.$$

Proof. Transport-state deformation is a special case of standing-preserving deformation. It changes how the class participates in bounded transport or restoration while retaining identity. The corresponding compression response is therefore evaluated by the mass-role profile. Restricting $\mu_D(\Pi)$ to this transport-state subfamily yields $\iota_D(\Pi)$. Thus mass is the general standing-preserving deformation-response profile, while inertia is its transport-facing manifestation. The inertial role is resistance of particle identity to admissible changes of transport state. No Newtonian law, force equation, or spacetime acceleration parameter is assumed. $\square$

7 Mass-inertia role induction

Theorem 7.1 (Mass-inertia role induction). *In a fixed admissible realized persistence regime with field-carrier mediation, particle persistence classes, gauge-compatible presentation, bounded perturbation transport, and quotient comparison, any identity-retaining deformation family induces a mass/inertia role:*

$$(\Phi_D, \Pi_D, \mathcal{G}_D, \text{BoundTrans}_D, C_D) \wedge \text{DefAdm}_D \wedge \text{IdRet}_D \Rightarrow \text{MassInertiaRole}_D.$$

Moreover,

$$\text{MassInertiaRole}_D = (\mu_D, \iota_D), \quad \mu_D(\Pi) : \Delta_\Pi \rightharpoonup C_D, \quad \iota_D(\Pi) = \mu_D(\Pi)|_{\Delta_\Pi^{\text{tr}}}.$$

Proof. Particle identity is supplied by Π_D , not by primitive objecthood. A deformation of a localized carrier configuration threatens the class unless the standing signature, boundary/classification witness, identity-memory, restoration profile, perturbation-burden profile, transport profile, quotient comparison, and gauge-compatible presentation remain class-preserving. If identity is retained, the deformation must be mediated by a standing-preserving compression response. That response is carried through field-carrier mediation, evaluated through restoration/transport structure, and compared through C_D . Gauge-equivalent presentations cannot change the physical response. The induced assignment of deformation witnesses to quotient-stable compression-response classes is μ_D . By the deformation-mediation proposition, this profile is operative when changes to it alter the deformation and transport behavior preserving Π_D . Restricting to transport-state deformations yields ι_D . Thus the minimal mass/inertia role is forced by identity-retaining deformation. $\square$

8 Mass-class witnesses

Proposition 8.1 (Mass-class witness condition). *A mass-class witness χ_U^m is admissible as a particle-class witness only if it preserves $\mu_D(\Pi)$ under admissible carrier transport, restoration, gauge transition, and boundary comparison.*

Proof. A particle-class witness distinguishes a persistence class from alternatives. A mass-class witness distinguishes classes by deformation response. If the witness fails under transport, restoration, gauge transition, or boundary comparison, it is not class-stable; it is a presentation artifact or local extraction record. If it preserves $\mu_D(\Pi)$ across those operations, it can serve as a genuine classification witness for the mass role. $\square$

9 Relation to standard mass mechanisms

The present paper does not derive numerical masses. It identifies the physical role that numerical masses and mass-generation mechanisms must occupy.

9.1 Higgs/Yukawa and Standard Model masses

In the Standard Model, fermion masses are tied to Yukawa couplings and the electroweak Higgs vacuum structure; gauge-boson masses arise through electroweak symmetry breaking; much hadronic mass is controlled by QCD dynamics; neutrino masses involve mass/flavor-basis structure beyond the minimal charged-fermion pattern. These mechanisms instantiate deformation-response stabilization, transport-state-change resistance, and persistence-class preservation in specific field and gauge sectors. They are not assumed here; the present result derives the role they later specialize. A Yukawa entry, Higgs vev, pole mass, running mass, or mass matrix is physically mass-relevant only insofar as it represents such a profile inside its proper envelope.

9.2 Running and effective masses

A running or effective mass is not primitive same-domain drift of a mass essence. It is a transport-defined or envelope-dependent representation of $\mu_D(\Pi)$ under scale, matching, scheme, or effective-domain change. This aligns the present result with the earlier standing-preserving scale-transport analysis: licensed running is admissible variation under transport law; unlicensed reassignment is selector import.

9.3 Inertial and gravitational mass

The present paper derives inertial role as transport-state deformation response. It does not derive gravitational mass, stress-energy coupling, or the equivalence principle. It prepares those later questions by fixing the profile to which any gravitational/stress-energy role must couple if inertial and gravitational roles are to coincide. The gravitational paper cannot simply import a stress-energy ledger; it must connect metric/gravity-facing structure to the same compression-response profile.

9.4 Energy

Mass is not identified with energy here. Energy-like quantities may later measure continuation capacity, deformation work, transport budget, excitation content, or stress-energy coupling. The mass role is more specific: it is resistance of localized particle identity to standing-preserving deformation. Relativistic mass-energy relations are later specializations, not assumptions of this paper.

10 No hidden mass reservoir

Corollary 10.1 (Hidden mass reservoir exclusion). *A latent mass property inaccessible to boundary traces, gauge-compatible presentation, deformation witnesses, and quotient comparison cannot carry realized mass role. If it becomes accessible, it is represented in μ_D . If it remains inaccessible, it is realization-inert.*

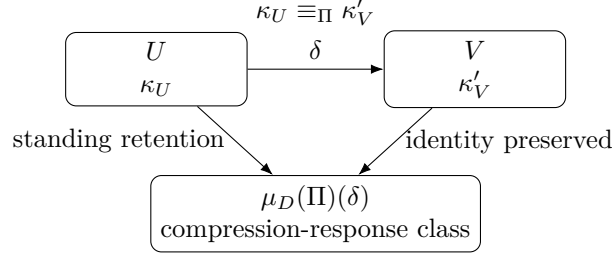
Proof. A realized mass role must affect the deformation-response organization of a particle persistence class. If a latent reservoir is inaccessible to all operations that fix that organization, it cannot alter or preserve standing-bearing deformation response. If it is accessible, it is no longer hidden and belongs to the admitted profile. Hence a hidden mass reservoir does not rescue primitive mass. $\square$

11 Finite witness model

A finite witness model makes the structure concrete without defining mass as graph weighting. Let

$$G_D = (V, E, \Phi, \Pi, \mathcal{G}, \Delta, \mu)$$

where V are local continuation domains, E are admitted adjacency/transport links, $\Phi(v)$ is a field-carrier profile, Π is a particle persistence class, $\mathcal{G}$ is gauge-role structure, Δ is a family of deformation witnesses, and μ is a compression-response profile.



The model separates collapse cases:

1. A primitive scalar m with no deformation role is skin.
2. A substance amount with no deformation response is unsupported.
3. A gauge-presentation difference counted as mass is a gauge artifact.
4. A deformation with no compression-response profile leaves identity unsupported.
5. An arbitrary compression-response value imports a selector.
6. A neutral response profile is admissible as a massless or neutral case relative to the deformation family.
7. A stable compression-response profile supplies the mass/inertia role.

The witness graph instantiates one admissible realization of deformation-response organization. It does not define mass as graph weighting.

12 Route exhaustion

Theorem 12.1 (Mass/inertia route exhaustion). *Any proposed primitive source of the mass/inertia role in a fixed admissible realized persistence regime reduces to one of the following:*

- (i) *deformation-response profile;*
- (ii) *primitive scalar mass assignment;*
- (iii) *primitive substance amount;*
- (iv) *energy ledger alone;*
- (v) *gauge-presentation artifact;*

- (vi) *hidden mass reservoir;*
- (vii) *measurement or extraction record alone;*
- (viii) *Standard Model parameter imported before role fixation;*
- (ix) *cross-domain imported value.*

Only (i) supplies the mass/inertia role without selector import, identity reset, gauge artifact, or domain drift.

Proof. A primitive scalar mass assignment lacks role unless it represents μ_D . A primitive substance amount lacks deformation-response content unless it reduces to μ_D . An energy ledger alone is too broad to identify the mass role. A gauge-presentation difference is not physical mass unless boundary traces distinguish the compression profile. A hidden reservoir is inaccessible or else represented in the profile. A measurement or extraction record fixes an observation of a mass role; it does not create the role. A Standard Model parameter imported before carrier, particle, gauge, and deformation structure are fixed is a later specialization used too early. A cross-domain value is not same-regime mass unless transported by admissible matching. Therefore the only surviving source is the deformation-response profile. $\square$

13 Normal form

The result can be compressed as

$$(\Phi_D, \Pi_D, \mathcal{G}_D, \text{BoundTrans}_D, C_D) \wedge \text{DefAdm}_D \wedge \text{IdRet}_D \Rightarrow \text{MassInertiaRole}_D$$

with

$$\text{MassInertiaRole}_D = (\mu_D, \iota_D), \quad \mu_D(\Pi) : \Delta_\Pi \rightarrow C_D, \quad \iota_D(\Pi) = \mu_D(\Pi)|_{\Delta_\Pi^{\text{tr}}}.$$

The physical interpretation is equally compact: mass is the operative deformation-response profile of particle identity; inertia is its transport-state-change restriction. Both are carried by field mediation, stabilized by particle persistence-class identity, fixed against gauge presentation artifacts, and compared through the quotient comparison carrier.

14 Conclusion

Paper 1 established that realized persistence cannot be physically null. Paper 2 established bounded localization. Paper 3 derived continuation-neighborhood domains. Paper 4 derived the metric role of quotient-stable comparison. Paper 5 derived bounded perturbation transport. Paper 6 derived fields as persistence carriers. Paper 7 derived particles as localized persistence classes. Paper 8 derived gauge structure as boundary-fixed presentation equivalence. The present paper shows that particle persistence under admissible deformation requires a stable compression-response profile.

What has been isolated is the minimal deformation-response role required for localized persistence-class identity under field-carrier mediation and bounded perturbation transport. Mass is the general standing-preserving response profile; inertia is its transport-state-change manifestation. The result does not supply numerical mass spectra, Higgs/Yukawa mechanisms, stress-energy coupling, gravitational mass, or relativistic mass-energy dynamics. It fixes the role those later structures must satisfy: familiar mass and inertial structures are specializations of a gauge-compatible, boundary-fixed, quotient-stable compression-response profile of a localized particle persistence class.

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